

NARRABRI, Australia

WMO: 957340

Lat: 30.3153S

Lon: 149.8303E

Elev: 753

StdP: 14.30

Time Zone: 10.00 (AUS)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.9	35.6	18.2	14.1	63.4	22.0	17.0	60.6	24.0	63.0	21.2	63.2	2.5	170	0.485

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.1	101.5	68.1	98.5	67.5	95.7	66.8	72.8	87.6	71.7	86.3	70.6	85.7	11.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.9	113.0	75.0	67.9	105.5	75.5	66.2	99.4	76.0	37.0	87.5	35.9	86.1	35.0	85.1	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	20.9	18.6	27.8	106.3	2.1	4.7	26.3	109.7	25.1	112.4	23.9	115.1	22.4	118.5
			26.4	75.5	2.2	2.2	24.8	77.1	23.5	78.4	22.3	79.6	20.7	81.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.9	81.6	79.5	74.7	67.4	58.9	53.7	52.0	54.0	60.5	68.3	73.9	79.1
	DBStd	11.87	6.10	5.84	5.13	5.24	5.10	4.67	4.83	5.38	6.07	6.49	6.68	6.13
	HDD50	75	0	0	0	0	3	20	33	18	1	0	0	0
	HDD65	1530	0	0	2	33	195	340	403	345	164	40	7	1
	CDD50	6243	980	826	767	523	279	130	95	142	315	567	716	903
	CDD65	2223	515	406	304	106	7	0	0	4	27	142	273	439
	CDH74	26355	6439	4848	3110	996	88	0	2	38	444	1682	3237	5471
	CDH80	12877	3562	2556	1317	238	4	0	0	5	117	616	1537	2925
(14) Wind	WSAvg	8.9	10.1	9.6	9.0	8.2	6.9	7.5	7.5	8.2	9.1	9.8	10.3	10.7
(15) Precipitation	PrecAvg	24.9	4.4	2.6	1.7	1.6	2.5	1.3	1.7	1.3	1.3	1.7	1.8	2.6
	PrecMax	35.5	11.7	8.7	5.4	6.9	8.2	3.9	6.2	5.0	4.1	5.6	4.1	6.5
	PrecMin	12.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1
	PrecStd	7.4	3.5	2.3	1.6	1.9	2.3	1.0	1.5	1.3	1.3	1.5	1.3	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	105.3	104.8	97.2	89.5	79.7	71.7	73.0	78.2	88.7	94.5	103.5	104.5
		MCWB	69.9	69.4	67.5	63.1	59.8	56.6	55.2	57.0	59.6	62.5	66.2	66.5
	2%	DB	101.1	100.0	93.5	85.5	76.4	68.8	69.2	73.8	83.3	90.2	96.8	99.4
		MCWB	69.2	68.0	66.6	63.3	58.6	55.6	54.4	55.7	58.7	62.4	64.8	66.6
	5%	DB	97.9	96.5	90.2	82.6	73.9	66.6	66.2	70.6	79.0	86.5	92.4	96.6
		MCWB	68.4	67.6	65.9	62.1	57.3	54.7	53.6	54.0	57.4	61.3	64.2	66.1
	10%	DB	94.8	92.7	87.3	79.9	71.5	64.0	63.5	67.5	75.2	83.0	88.4	93.4
		MCWB	68.0	67.0	65.6	61.1	56.4	53.8	52.7	52.9	56.4	60.8	63.6	65.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.9	75.1	72.3	68.7	64.2	61.0	62.0	61.7	64.6	68.2	71.2	72.8
		MCDB	89.5	89.7	84.2	78.5	71.3	64.1	66.1	74.0	78.0	82.1	86.8	85.2
	2%	WB	73.0	72.8	70.8	66.2	61.7	58.6	57.9	58.7	62.4	65.8	69.1	71.0
		MCDB	87.3	89.1	82.6	78.8	71.2	64.0	64.8	66.5	74.2	80.7	84.5	85.3
	5%	WB	71.8	71.3	69.4	64.2	59.7	56.9	55.3	56.4	60.5	64.1	67.7	69.8
		MCDB	87.2	86.2	81.9	76.5	70.1	63.6	62.9	67.0	72.3	79.6	82.5	85.4
	10%	WB	70.8	69.9	68.0	62.8	57.8	55.3	53.4	54.3	58.4	62.7	66.4	68.7
		MCDB	86.8	85.7	81.5	76.1	68.9	62.4	62.3	65.4	71.9	78.0	81.7	84.6
(35) Mean Daily Temperature Range	5% DB	MDBR	23.1	23.7	23.0	24.2	24.6	20.5	22.9	25.6	27.3	26.1	24.8	24.9
		MCDBR	25.8	27.7	26.3	26.4	26.5	23.3	26.0	29.5	32.1	30.2	29.9	27.6
		MCWBR	8.2	8.6	9.5	10.6	12.5	12.4	14.5	15.4	14.8	12.5	10.6	9.8
	5% WB	MCDBR	21.5	22.4	21.0	22.2	21.4	17.9	20.8	23.0	24.9	24.5	22.4	22.4
		MCWBR	7.6	7.9	8.5	9.5	10.6	10.6	12.7	13.1	13.3	11.1	8.9	8.9
(40) Clear-Sky Solar Irradiance	taub		0.360	0.361	0.334	0.315	0.296	0.290	0.285	0.297	0.330	0.349	0.373	0.361
	taud		2.532	2.540	2.609	2.618	2.616	2.641	2.637	2.575	2.474	2.458	2.439	2.504
	Ebn at Noon		311	305	303	293	285	279	286	295	299	305	305	311
	Edh at Noon		35	34	30	27	24	22	23	27	34	36	38	36
(44) All-Sky Solar Radiation	RadAvg		2356	2140	1823	1502	1171	943	1069	1372	1759	2060	2213	2361
	RadStd		143	156	118	79	63	78	68	81	126	159	219	175

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		+1.04	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+409	+270	

Nomenclature: See separate page